Economic transformation and environmental pressure in global production networks

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Abstract

Changes in sectoral shares do not fully describe structural change in a production network. We examine the distribution of economic value, environmental sources and production paths in EXIOBASE 3.10.2, retaining 163 industries and 49 regions throughout. Economic accounts cover 1995–2024; the common environmental panel covers 1995–2022. Absorbing-path calculations distinguish expected intermediate stages, path dispersion, branching and repeated regional crossings. Exact decompositions separate changes in originating source weights from changes in their production relationships. Exploratory principal component analysis and clustering compare four economic and four environmental characteristics across 1,232 country-year observations. The value-added share generated for foreign final demand rose from 15.57% to 22.16%, while manufacturing's share declined and the broad service share changed modestly. Selected carbon dioxide and used material inventories increased by 61.8% and 90.6%, respectively. Income-weighted paths lengthened, but length and branching followed different histories: material paths showed little length change through 2019 despite increased branching per decision. Two principal components explain 75.5% of country-profile variation. Cluster overlap and sensitivity to country resampling favour continuous profiles over fixed country types. The findings distinguish economic diversification from environmental diversification and identify production-network changes concealed by average length or final-demand geography alone. They describe accounting relationships and do not identify causal trade effects or feasible policy substitutions.

1. Introduction

Global structural change concerns both the composition of production and the relationships through which production takes place. Changes in national income shares, manufacturing and services describe the first dimension. Relationships among suppliers, intermediate users and final purchasers describe the second. These dimensions can move differently. Services can gain a larger share of value added while depending on expanding material production. An industry can sell mainly to domestic customers while its products enter supply chains that cross borders several times. Economic transformation therefore requires an account of production networks alongside an account of sectoral shares.

The environmental consequences depend on both dimensions. Emissions and extraction originate in particular industries and places, whereas the purchases sustaining those activities may occur elsewhere. Concentration at the point of production need not imply concentration among final purchasers. Conversely, a geographically dispersed economy can depend on a narrow set of emitting activities. Comparing economic and environmental structures requires separate measures of the amount produced, its distribution across sources, and the production relationships connecting those sources to final demand. A change in any one of these measures does not determine the others.

This paper asks: **How did the distribution of economic value and the organisation of global production change between 1995 and 2024, and how did those changes relate to environmental pressure?** We examine three aspects of production networks: the number of intermediate stages separating an originating industry from final demand; the diversity of possible paths between them; and the frequency of crossings between regions. We then use exploratory machine learning to identify recurring combinations of economic structure, production-network characteristics and environmental dependence across countries. This combination makes the economic and environmental dimensions comparable without reducing them to a single index.

Wood and Lenzen (2009) distinguish several aggregate descriptions of economic structure and show why economic diversification need not imply environmental diversification. Their review motivates our use of complementary measures. We extend the empirical scope to a global annual network and give explicit attention to the distinction between concentration across industries and complexity within production paths. A distribution can become more concentrated even while the paths connecting its sources to final demand become more numerous. Neither development alone establishes an improvement or deterioration in economic performance.

Two established approaches provide the analytical foundations. Environmentally extended input–output accounting connects pressure recorded at production sources to the final purchases supported by that production. Research on emissions embodied in trade and material footprints demonstrates the importance of this source–destination distinction (Peters et al., 2011; Wiedmann et al., 2015). Production-length and absorbing-Markov-chain approaches describe how activity passes through intermediate stages before reaching final use (Antràs et al., 2012; Dietzenbacher and Romero, 2007; Kostoska et al., 2020). We use both perspectives to examine value added as well as environmental pressure.

Average length alone leaves important features unresolved. Two networks can have the same mean number of stages but very different distributions around that mean. They can also differ in how many routes are possible and how often a route crosses a regional boundary. A chain that returns to its region of origin may have substantial international exposure even though its final purchaser is domestic. We therefore distinguish final-demand destinations from border-crossing histories. The distinction is relevant when interpreting changes in trade-related environmental pressure as evidence about the organisation of production.

The machine-learning analysis has a narrower purpose. It identifies similarities among observed country-year profiles and describes how those profiles change. It does not estimate causal effects or discover objectively correct country types. Principal component analysis makes the dominant combinations of measured characteristics inspectable; clustering tests whether useful groupings can summarise the same information. Stability exercises preserve country histories and examine changes in scaling and time coverage. These choices address the dependence among annual observations rather than treating each country-year as an independent experiment.

The contribution is an integrated comparison of economic value creation, environmental sources and production-path characteristics at a fixed, detailed input–output resolution. The path measures use established Markov and information-theoretic identities. The decompositions are exact accounting identities with explicit conventions for interaction terms. The novelty sought here is empirical and interpretive: identifying which dimensions of global structural change move together, which diverge, and which apparent patterns depend on the period or statistical representation. Claims about feasible policies require additional assumptions and are kept separate from the historical account.

2. Data, boundaries and modelling choices

2.1 Database and period

We use the industry-by-industry tables in EXIOBASE 3.10.2, an immutable release identified by DOI 10.5281/zenodo.20051562. Each annual table contains 163 industries in 49 regions, or 7,987 region–industry accounts. The economic series comprises thirty annual observations, 1995–2024. This is approximately three decades: twenty-nine elapsed years between the endpoints. One release is used throughout because combining vintages would confound economic change with revisions to source data and balancing procedures. Checksums identify every downloaded archive.

EXIOBASE is selected for its combination of industrial detail, long annual coverage and extensive environmental accounts. OECD ICIO provides a strong alternative for international economic relationships and covers more countries separately. GLORIA and Eora offer broader geographical coverage, while WIOD's previous-year-price tables are valuable for volume-based decomposition. The choice here prioritises a common detailed production system for several environmental indicators and value added. It does not imply that EXIOBASE is preferable for every question. Independent database replication remains outside the executed analysis; agreement with a separate global carbon inventory is a more limited check.

The common environmental panel ends in 2022. Direct inspection of the later archives finds absent water, energy and employment files, an entirely zero land account, and zero essential combustion rows in the air-emission accounts. Partial process-emission entries cannot substitute for complete carbon or methane inventories. These indicator-years are therefore excluded. Materials remain usable through 2024, although their later estimates and unresolved allocation require additional caution. File presence, calendar coverage and empirical completeness are treated as separate properties. The same availability rules apply before any trend or machine-learning calculation.

The 49-region classification is fixed throughout. Forty-four regions are named countries and five combine the remaining economies. A crossing between two countries within one combined region cannot be identified as a regional crossing. We retain all regions and industries in every production-system solve. For the country-profile analysis alone, the five combined regions are excluded as observations because they are not comparable country units; their production relationships remain in the global network supplying the features. This changes the observational sample for machine learning, not the input–output system.

2.2 Economic and environmental accounts

The principal economic quantity is industry value added, gross of depreciation: compensation of employees, operating surplus and other net production taxes. Taxes less subsidies on products purchased are excluded from industry value added but retained in the full column-balance check. Value added measures income generated through production and avoids the repeated counting inherent in gross output. Gross output and intermediate transactions are informative about the scale of interindustry exchange. We report both, with explicit denominators. Monetary accounts are in current-price millions of euros; their changes reflect prices and exchange rates as well as quantities.

Economic composition is described by countries' shares of world value added, the shares generated in manufacturing and services, and the concentration of value added across industries and regions. Manufacturing comprises the detailed manufacturing industries; services comprise trade, hospitality, transport, communications, business services, public services and other services. Energy, water, waste treatment and construction are reported separately rather than folded into an expansive service category. These classifications organise presentation and features; they never replace the detailed production coefficients. Exact industry membership is documented in the appendix and code.

The climate headline is fossil and process carbon dioxide, combining combustion, cement-process, lime-process and fossil-waste rows. Biogenic carbon dioxide and the separately recorded peat-decay row are excluded. Methane is retained separately in physical mass. This avoids combining gases with undocumented characterisation factors, but the carbon account is consequently not a complete greenhouse-gas footprint. Materials comprise used domestic extraction of biomass, fossil fuels, ores and non-metallic minerals. Adding unused extraction would answer a different question about disturbance associated with extraction rather than material entering use.

Land is the area recorded in the land account, including forests and pastures. Blue water is consumption from surface-water and groundwater sources, excluding withdrawals and green water. Final energy uses the explicitly named final-energy account; overlapping gross and net energy rows are not added. Employment counts people across the six sex-and-skill categories, not hours. These choices retain physical comparability within each indicator. They do not make the indicators interchangeable measures of environmental damage: water scarcity, ecological condition, extraction impacts and employment quality require additional information.

2.3 Inventories and their destinations

For an environmental indicator, some pressure is recorded in industries with positive monetary output, some is released directly by final users, and some is attached to accounts with zero output. We call the first quantity the **industrial inventory that can be traced through the monetary production system**. The shorter term *traceable industrial inventory* always refers to this quantity. Direct final-user pressure includes, for example, household fuel combustion. Zero-output residuals have no defined pressure-per-output coefficient and therefore cannot be traced to final purchasers using the standard monetary model.

All three components enter worldwide inventory totals. Only the traceable industrial component enters industrial production-path measures. This preserves reported pressure without inventing connections for zero-output accounts. We do not divide by an arbitrary small number or silently discard the residual. For material extraction, the residual becomes sufficiently large near the end of the series to qualify small changes in destination shares. Bounds place all residual pressure first with final demand in the source region and then with final demand elsewhere; these are allocation bounds, not statistical confidence intervals.

Throughout the paper, attribution runs **from an originating industry and region to the final demand whose production requirements include that industry's output**. For carbon, the source is where emissions occur; for materials, where extraction occurs; for value added, where income is generated. The destination is household, government, investment or other final demand, located in a specified region. A footprint attributed to a destination sums the upstream sources required to satisfy its purchases under the model. Attribution establishes an accounting relationship, not exclusive moral responsibility or a causal effect of the purchaser.

Signed final demand is retained in the historical accounting system because inventory withdrawals and other negative entries are part of the published balance. Probabilistic path measures require nonnegative transitions and final-use probabilities. For those measures we use a separate positive-final-demand closure, defined below, retaining the original input coefficients. The historical inventory totals remain unchanged. Comparing these two constructions identifies the effect of the signed-demand convention without presenting the diagnostic closure as another observed economy.

3. Methods

3.1 Input–output accounts and notation

For a given year, let i index a supplying region–industry account, j a using account, and d a final-demand destination region. The function $r(i)$ returns account i's source region. There are n active accounts and $R = 49$ regions. Bold lowercase letters denote column vectors, bold uppercase letters matrices, and a superscript $\top$ transposition. The vector $\mathbf{1}$ contains ones of the dimension required by the expression; $\mathbf{I}$ is the identity matrix. A hat denotes a diagonal matrix formed from a vector.

The element Z_{ij} is the intermediate delivery from supplier i to user j, x_i is gross output, and Y_{id} is final demand for account i's output in destination d, summed over expenditure categories. Total final demand is $\mathbf{y} = \mathbf{Y1}$. Direct input coefficients and the Leontief inverse are

$$\mathbf{A} = \mathbf{Z}\widehat{\mathbf{x}}^{-1}, \qquad \mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}, \qquad \mathbf{x} = \mathbf{Ly}.$$

For indicator a, let $e_i^{(a)}$ be its traceable industrial inventory at account i, $E^{(a)} = \sum_i e_i^{(a)}$ the global total, and $f_i^{(a)} = e_i^{(a)}/x_i$ its direct intensity. With $\mathbf{X} = \mathbf{LY}$, the industrial footprint attributed from all source accounts to destination d is

$$F_d^{(a)} = \sum_i f_i^{(a)} X_{id}, \qquad \sum_d F_d^{(a)} = E^{(a)}.$$

Direct final-user pressure is added to the relevant destination separately. Denote its worldwide total by $D^{(a)}$ and the zero-output residual by $O^{(a)}$. The complete reported inventory is $T^{(a)} = E^{(a)} + D^{(a)} + O^{(a)}$. For value added, replace $e_i^{(a)}$ by the signed value added v_i and define $V = \sum_i v_i$. The coefficient v_i/x_i then traces income generated at source account i to final demand in each destination. The corresponding destination totals sum to V.

The model assumes proportional input requirements and a common product mix within each industry, with final demand specified outside the production system. These assumptions make the observed accounts traceable; they do not describe substitution, capacity limits or price adjustment. Industry-by-industry tables align income and environmental sources with producing activities. Capital formation remains final demand because the annual tables do not identify the subsequent services and operating requirements of each asset.

We solve linear systems rather than explicitly form the full inverse. Labels, balance identities and reconstructed output are checked annually. Zero-output accounts are excluded from a solve only after their monetary rows, columns and final-demand entries are verified to be zero. This retains every economically active link without adding arbitrary thresholds. Double precision is used; componentwise refinement and a separately scaled transition solve protect probability calculations for extremely small positive-output accounts. Numerical tolerances and their observed maxima are reported in the audit appendix.

3.2 Source distributions and final-demand destinations

For any nonnegative source quantity, let w_i be account i's share of its worldwide total, so that $\sum_i w_i = 1$. Environmental weights are $w_i^{(a)} = e_i^{(a)}/E^{(a)}$. Probabilistic economic summaries use $w_i^{(v)} = v_i^+/V^+$, where $v_i^+ = \max(v_i, 0)$ and $V^+ = \sum_i v_i^+$. Negative value added is retained in monetary totals and destination accounting; its absolute share is audited. Positive weights are required only when interpreting a mixture as a probability distribution. Gross-output weights provide a secondary economic description with their repeated-counting limitation stated.

Concentration across a specified partition is measured by the effective number of equally weighted categories,

$$N_{\text{eff}}(\mathbf{s}) = \exp\left(-\sum_{c:s_c>0} s_c \ln s_c\right),$$

where c indexes a country or industry and s_c is its share after summing the relevant account weights. A decrease means that the source quantity is distributed less evenly. It does not mean industries disappeared or that feasible substitutes became scarcer. We use the same measure for value added and environmental pressure to make concentration comparable, while retaining separate physical and monetary totals. A more even distribution can arise through reductions in large sources or increases in small ones; concentration has no intrinsic environmental sign.

For source account i, define the share of its output required by final demand outside its source region as

$$q_i = 1 - \frac{X_{i,r(i)}}{x_i}, \qquad Q(\mathbf{w}) = \sum_i w_i q_i.$$

Thus Q is the fraction of the selected source quantity attributed from its originating region–industry accounts to final demand in another region. We use the phrase *share serving final demand abroad* for this measure. For an environmental inventory, its physical amount is $W^{(a)} = E^{(a)} Q(\mathbf{w}^{(a)})$. For value added, signed monetary attribution uses v_i directly. An emission or unit of value added is counted once by ultimate destination, irrespective of how many intermediate borders its associated production path crosses.

The historical mean number of intermediate stages associated with source account i is

$$u_i = \frac{(\mathbf{L}\mathbf{x})_i}{x_i} - 1, \qquad m(\mathbf{w}) = \sum_i w_i u_i.$$

This is the mean of the signed production-layer accounting contributions. Stage zero refers to a source industry producing directly for final demand; each preceding intermediate delivery adds a stage. The statistic counts accounting transitions, including repeated transitions in cycles, rather than kilometres, establishments or elapsed time. Because final demand can be negative, probability statements below use a separate nonnegative construction. The distinction replaces ambiguous descriptions of an industry's downstream position with an explicit quantity and boundary.

3.3 Production paths, branching and repeated borders

Let $\mathbf{Y}^+$ sum the positive parts of the original detailed final-demand cells into destinations. Taking the positive part before summing prevents withdrawals in one expenditure category from cancelling purchases in another. Define $\mathbf{y}^+ = \mathbf{Y}^+\mathbf{1}$ and the output required by this demand as $\mathbf{x}^+ = \mathbf{L}\mathbf{y}^+$. The original input coefficients are retained. Intermediate deliveries in this closure are $\mathbf{A}\widehat{\mathbf{x}}^+$, so the construction is internally consistent rather than a row-normalisation of an unbalanced table.

The downstream transition matrix $\mathbf{P}$ and final-use probability matrix $\mathbf{B}$ are

$$P_{ij} = \frac{A_{ij} x_j^+}{x_i^+}, \qquad B_{id} = \frac{Y_{id}^+}{x_i^+}, \qquad \mathbf{P}\mathbf{1} + \mathbf{B}\mathbf{1} = \mathbf{1}.$$

Starting at source account i, a path either moves to an intermediate user j or terminates in final demand at destination d. The probabilities reflect shares of output disposition in this diagnostic closure. They are a representation of the accounting network, not observations of randomly routed physical products. The Markov assumption applies the same industry's average disposition whenever the path visits that industry. It is justified by the resolution of the available accounts; it does not imply that individual firms choose customers independently.

Let $\mathbf{N} = (\mathbf{I} - \mathbf{P})^{-1}$ be the fundamental matrix, whose element N_{ij} is the expected number of visits to account j starting at i, including the initial visit. Let K_i be the number of intermediate transitions before final use, with mean μ_i and second moment τ_i. Then

$$\boldsymbol{\mu} = \mathbf{N}\mathbf{1} - \mathbf{1}, \qquad \boldsymbol{\tau} = 2\mathbf{N}\boldsymbol{\mu} - \boldsymbol{\mu}, \qquad \sigma^2(\mathbf{w}) = \sum_i w_i \tau_i - \left(\sum_i w_i \mu_i\right)^2.$$

The variance σ^2 includes variation among paths from the same source and variation among starting sources. It measures dispersion of accounting lengths, not the probability of disruption. Reporting it alongside the mean addresses the possibility that equal averages conceal different path distributions. All moments are obtained by solves; no arbitrary maximum chain length truncates them.

To measure branching, define the one-decision entropy h_i and the entropy H_i of the complete path conditional on its starting account:

$$h_i = -\sum_j P_{ij} \log_2 P_{ij} - \sum_d B_{id} \log_2 B_{id}, \qquad \mathbf{H} = \mathbf{N}\mathbf{h}.$$

Terms with zero probability contribute zero. Entropy is in bits. A deterministic route has zero entropy even if it contains several stages. More alternative paths, or more even use of existing alternatives, increase entropy. The recursion $\mathbf{H} = \mathbf{h} + \mathbf{P}\mathbf{H}$ sums the uncertainty encountered at every visit, including repeated visits in cycles. This applies the information-theoretic treatment of Markov trajectories to production paths (Ekroot and Cover, 1993). No claim is made that greater entropy measures technological sophistication or economic welfare.

Since longer paths offer more opportunities to branch, we also report

$$\beta(\mathbf{w}) = \frac{\sum_i w_i H_i}{1 + \sum_i w_i \mu_i}.$$

The denominator counts expected decisions, including final absorption. Thus β is path entropy per expected decision, not a stationary entropy rate. It distinguishes increases in total path entropy associated mainly with more stages from increases associated with greater branching at each decision. Actual source weights are retained when comparing carbon, materials and value added, so differences reflect their positions within the same diagnostic network rather than changes in the observed inventory totals.

Finally, let c_i be the probability that the next transition crosses a regional boundary:

$$c_i = \sum_j P_{ij} 1\{r(i) \neq r(j)\} + \sum_d B_{id} 1\{r(i) \neq d\}, \qquad \mathbf{b} = \mathbf{N}\mathbf{c}.$$

Here $1\{\cdot\}$ equals one when its condition is true and zero otherwise; b_i is the expected number of regional crossings, including the last delivery to final demand. We also compute a_i, the probability of at least one crossing, by subtracting from one the probability of eventual domestic final use along paths confined to the source region. The absorption matrix $\mathbf{G} = \mathbf{N}\mathbf{B}$ gives $q_i^+ = 1 - G_{i,r(i)}$, the probability of final use abroad. Consequently, $a_i - q_i^+$ is the probability of crossing a border and ultimately returning to domestic final demand. These quantities distinguish repeated international production from the final location of demand.

3.4 Decomposing changes in source weights and destinations

We separate changes in a weighted structural average into changes in its source weights and changes in the source accounts' measured relationships with final demand. *Source composition* means the distribution of value added or an environmental inventory across originating region–industry accounts. *Routing* means the account-level destination share q_i or stage measure u_i; it combines intermediate production relationships and final-demand patterns. It is not a synonym for transport routing or pure technical change. These definitions determine the interpretation of each component.

Let $t = 0, 1$ denote two dates, z_i stand for either q_i or u_i, and C contain accounts active at both dates. For any variable, Δ denotes the later value minus the earlier value and an overbar their arithmetic mean. The exact decomposition is

$$\Delta\left(\sum_i w_i z_i\right) = \underbrace{\sum_{i\in C} \overline{z}_i \Delta w_i}_{\text{source composition}} + \underbrace{\sum_{i\in C} \overline{w}_i \Delta z_i}_{\text{routing}} + \underbrace{\sum_{i\in E} w_{i1} z_{i1} - \sum_{i\in X} w_{i0} z_{i0}}_{\text{account entry and exit}}.$$

The sets E and X contain entering and exiting active accounts; they are distinct from an environmental total $E^{(a)}$. The symmetric convention divides the common-support interaction equally. Entry and exit are separate because an output-normalised property is undefined when output is zero. For physical pressure attributed to foreign final demand, we first decompose $\Delta W^{(a)} = \overline{Q}\Delta E^{(a)} + \overline{E}^{(a)}\Delta Q$, then divide the share contribution using the expression above. This hierarchy separates worldwide inventory scale from its destination distribution. Annual chained and endpoint contributions disclose dependence on the chosen comparison convention.

A positive diagonal revaluation of monetary units, $\mathbf{D}$, transforms $\mathbf{A}$ into $\mathbf{DAD}^{-1}$, output into $\mathbf{Dx}$ and final demand into $\mathbf{DY}$. Environmental footprints and account-level path properties are invariant under this consistent transformation. Monetary source weights generally are not. We therefore distinguish invariant network relationships from current-price economic composition and from environmental pressure per euro. This result does not remove buyer-specific price differences or recover physical technologies within heterogeneous industries. Conventional real-volume structural decomposition would require additional price information (Xu and Dietzenbacher, 2014).

3.5 Exploratory country profiles

The exploratory sample contains the 44 named countries in every year from 1995 through 2022. We select four economic features and four environmental features before fitting the models. Economic features are the service share of positive value added, the signed value-added share serving final demand abroad, value-added-weighted expected production stages, and value-added-weighted branching entropy per decision. Environmental features are the carbon and material shares serving final demand abroad and their relative production intensities. This gives equal total marginal variance to the two domains while including economic composition and network organisation explicitly.

For country r, relative intensity for indicator a is

$$I_r^{(a)} = \frac{E_r^{(a)}/E^{(a)}}{V_r/V},$$

where $E_r^{(a)}$ is traceable industrial pressure originating in r and V_r its signed industry value added. An intensity above one means that the country's share of the pressure exceeds its share of economic value. The logarithm is used because proportional differences are more comparable than absolute differences for these positive, skewed ratios. The denominator remains current-price value added; this feature is not a physical efficiency estimate. We do not use country size, year or future observations as target labels because the purpose is to compare observed functional profiles, not predict an outcome.

Each feature is centred on its pooled sample mean and divided by its pooled standard deviation. For observation o and feature f, let M_{of} be the resulting standardised matrix. Principal component analysis uses the singular-value decomposition $\mathbf{M} = \mathbf{U\Sigma V}^\top$. $\mathbf{U}$ contains left singular vectors and $\mathbf{\Sigma}$ is the diagonal matrix of singular values. Columns of $\mathbf{V}$ are loading vectors defining linear combinations of the features; $\mathbf{MV}$ contains the component scores, and squared singular values determine explained-variance shares. These matrices are unrelated to the Leontief notation or the scalar total value added V above. The loading sign is fixed for reproducible display but has no substantive meaning (Jolliffe and Cadima, 2016).

Clustering uses k-means in all eight standardised features, rather than only the first two plotted components. Candidate values are $k = 2, \dots, 6$, with fifty initialisations for each. We select the highest silhouette score among solutions in which every cluster contains at least five percent of observations. The silhouette compares within-cluster distances with distances to the nearest other cluster; larger values indicate clearer separation (Rousseeuw, 1987). The size restriction prevents a tiny outlying group from becoming the principal typology. Neither criterion establishes that discrete country types exist.

Sensitivity checks resample complete country histories sixty times, omit six consecutive multi-year blocks in turn, and substitute median/interquartile-range scaling. Agreement with the baseline grouping is measured by the adjusted Rand index, which equals one for identical partitions and has expected value near zero for random agreement under its reference model. We also fit PCA to within-country annual feature changes. A circular-shift reference independently shifts each feature's time series within a country, preserving its marginal distribution and temporal sequence while changing alignment with other features. Implementation uses scikit-learn (Pedregosa et al., 2011). These checks diagnose dependence and fragility; they do not manufacture independent observations or supply causal significance tests.

3.6 Interpretation and presentation

All reported environmental growth rates use complete selected inventories, whereas production-path and destination measures use their traceable industrial components. Economic income totals retain signed value added; probability-weighted economic paths use positive value added with the excluded negative share disclosed. Historical shares and positive-closure path probabilities are distinguished in text, captions and output names. These conventions follow the quantities each method requires rather than a preferred result. The appendix records every transformation and validation gate.

Figures use aligned panels, explicit units, restrained grids and a consistent colour-and-line vocabulary. Diverging scales are centred on a meaningful zero; colour is supported by legends, labels or line patterns. The selection follows Tufte's emphasis on quantitative comparison and contemporary guidance on accessible colour (Tufte, 2001; Crameri et al., 2020). Text describes the main patterns and their interpretation; complete annual values, loadings, cluster assignments and sensitivity results remain available in the downloadable outputs.

4. Results

4.1 Economic geography changed more than the global service share

The distribution of economic value shifted substantially towards China and India. China's share of world industry value added rose from 2.35% in 1995 to 17.81% in 2022, while India's rose from 1.19% to 3.44%. The United States retained approximately one quarter of the total. Japan's share fell sharply, although exchange rates and relative prices are important to interpreting its current-price comparison. These changes establish a redistribution of recorded economic value; they do not separately identify changes in physical output, productivity or purchasing power.

At the global level, manufacturing's value-added share declined from 22.25% to 16.73%. The service share increased much less, from 64.51% to 65.99%. This difference matters: falling manufacturing share did not translate proportionately into a rising service share. Extraction, energy, construction and other activities also occupy part of the economy. Within services, expanding business and financial activities coexist with declining shares in other groups. A description of global transformation as a uniform movement from manufacturing into services would therefore miss important changes within and outside both categories.

Economic concentration changed modestly over the full period but followed a non-monotonic geographical path. The effective number of value-added industries increased from 43.03 to 44.63. The effective number of source regions rose from 17.12 in 1995 to 21.77 in 2008, then returned to 17.49 by 2022. Thus the gains of emerging economies initially accompanied a more even distribution of income generation, but the endpoint is not substantially more geographically diverse than the starting point. Industry diversity and geographical diversity describe different aspects of the change.

Intermediate transactions increased as a share of gross output, indicating a larger monetary role for interindustry exchange. Their international share rose relative to 1995 but remained below its 2008 level. Neither statistic by itself describes real production fragmentation, because relative prices affect monetary flows. The value-added and path analyses below ask more specific questions: where income was generated for final demand and how many intermediate relationships connected its source industries to that demand.

The environmental contrast is pronounced. Selected worldwide carbon dioxide increased by 61.8% and used material extraction by 90.6% between 1995 and 2022. Their source distributions became less even across both industries and regions, whereas value-added industry diversity increased slightly. Blue water did not follow the same concentration pattern, and land area changed little in total. Figure 1 shows these principal differences; the complete annual accounts establish that economic diversification and environmental specialisation can coexist without either describing every indicator.

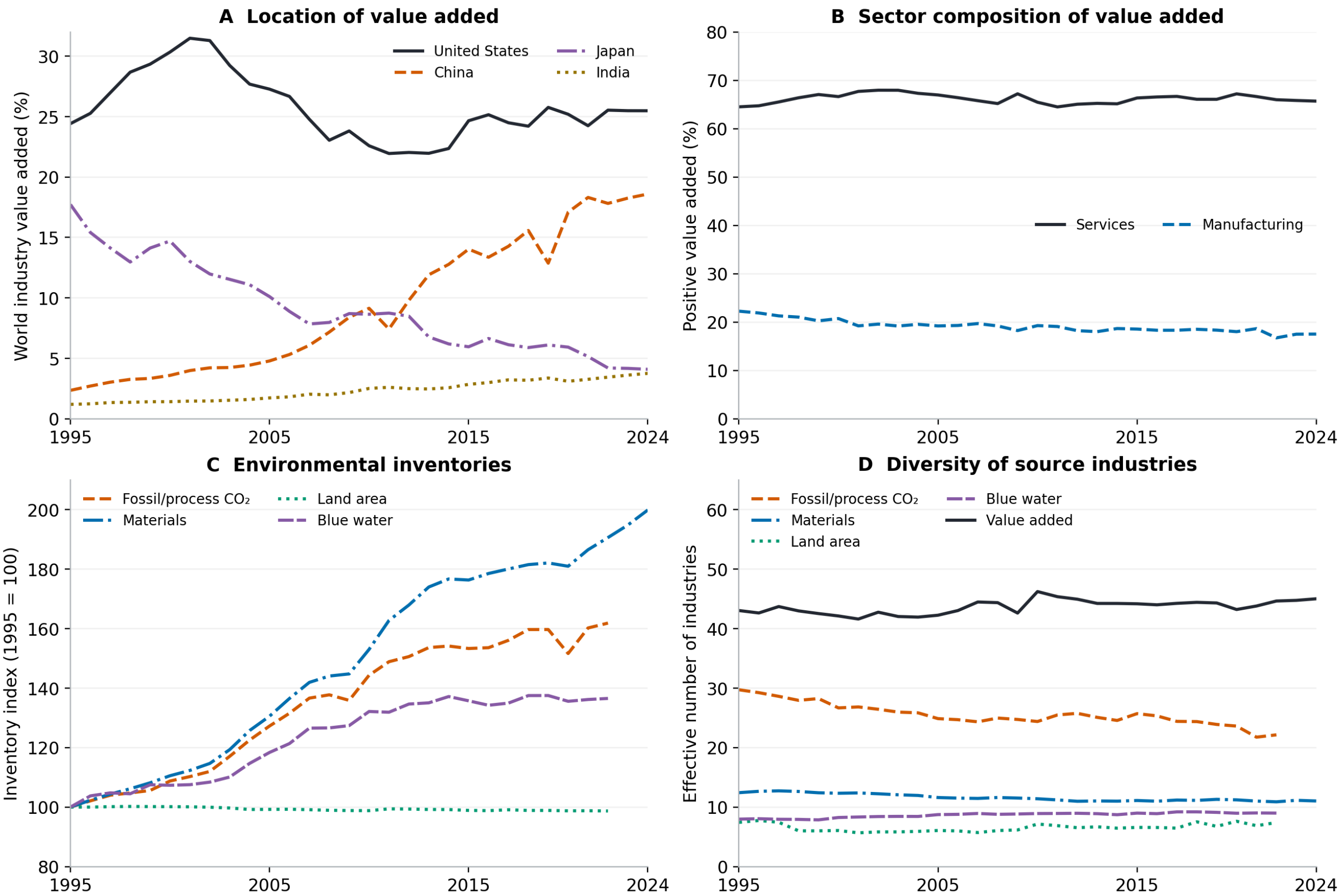

Figure 1. Economic composition and environmental pressure, 1995–2024. A: countries' shares of world industry value added. B: manufacturing and service shares of positive value added. C: complete selected environmental inventories, indexed to 1995. D: effective numbers of source industries for value added and environmental pressure, with an explicit panel legend. Monetary series are at current prices. Environmental carbon, land and water series end in 2022; only usable material and economic series continue.

4.2 Foreign final demand became more important for income and resources

The share of world value added generated for final demand outside its source region rose from 15.57% to 22.16%. Unlike the international share of intermediate monetary transactions, the 2022 value-added share exceeded its 2008 benchmark. This indicates that the location of final expenditure became more consequential for the geography of income generation, even though a simple gross-trade ratio did not rise continuously. Value-added accounting removes repeated counting of intermediate transaction values and assigns each source's income to its final-demand destinations.

The employment share associated with final demand abroad rose less, from 19.72% to 21.95%. Income and employment therefore converged in their aggregate destination shares while following different histories. This is consistent with changes in the distribution of remuneration, production activities and employment intensity, but the accounts do not identify those mechanisms separately. People counts also omit working hours, pay distribution and job quality. Their role here is to establish that the international organisation of income generation cannot be inferred from employment shares alone.

Environmental pressure had greater exposure to foreign final demand than either economic measure by 2022. The corresponding shares were 32.00% for traceable industrial carbon dioxide and 40.15% for material extraction. For both pressures, these shares were approximately 24% in 1995. The physical amounts attributed from producing regions to foreign final demand increased more strongly than the shares because worldwide inventories also expanded. Carbon attributed abroad more than doubled, while material extraction attributed abroad more than tripled.

These results concern the destination of final demand. They do not mean that the same fraction of physical products was exported, nor that trade caused the increase in global inventories. A domestic final purchase can require foreign income generation and environmental pressure through imported inputs. Conversely, an exported intermediate can eventually return within a product purchased domestically. The path calculations identify this second possibility explicitly instead of equating final-demand geography with every international production relationship.

4.3 Production paths became longer and more diverse, with different timing

Income generation moved further from final demand in the positive-demand path model. The value-added-weighted expected number of intermediate stages increased from 0.773 in 1995 to 1.082 in 2022 and 1.123 in 2024. Carbon- and material-weighted paths remained longer: their 2022 means were 1.919 and 2.350 stages. The contrast locates income generation closer to final purchases than resource extraction and carbon-emitting production on average. It does not imply that income or pressure physically travels between industries.

Dispersion also increased. The standard deviation of income-weighted path length rose from 1.303 to 1.634 stages; carbon and material paths had still larger dispersion in 2022. A mean around one or two stages therefore represents a broad distribution, including direct final production and substantially longer chains. These standard deviations describe variation among accounting paths and starting sources. They are not error bars around the estimated mean or estimates of disruption risk.

Complete-path entropy increased for all three quantities. Income-weighted entropy rose from 4.81 to 6.30 bits, carbon-weighted entropy from 8.65 to 10.03 bits, and material-weighted entropy from 8.18 to 11.12 bits. The material increase combines longer paths with greater branching per decision. For income and carbon, the distinction between these dimensions is especially important near the endpoint: branching per expected decision was higher in 2019 than in 2022, even though mean path length increased. The additional late stages did not coincide with an increase in every measure of complexity.

This difference in timing qualifies an interpretation based only on average production length. By 2019, material path length was little changed from 1995, but branching per expected decision had increased substantially. A nearly stable length could therefore conceal a more diverse set of production routes. After 2019, length increased while the branching measure receded. Figure 2 separates these movements rather than combining them into an index whose direction would depend on arbitrary weights.

Repeated regional crossings also became more important. In 2022, expected crossings were 0.289 per income-weighted path, 0.436 per carbon-weighted path and 0.559 per material-weighted path. The corresponding probabilities of any crossing were lower because a path can cross more than once. Among paths crossing at least one regional boundary, the expected counts were approximately 1.28, 1.33 and 1.36 respectively. The excess above one measures repeated crossings under the model's path distribution.

Paths that left and returned to domestic final demand accounted for less than half a percentage point of income-weighted paths, approximately 0.81 percentage points of carbon-weighted paths and 1.11 percentage points of material-weighted paths in 2022. These return probabilities are small relative to final demand abroad, but their existence is conceptually important. A destination-based account can classify a source quantity as domestically supported while the associated production path still depends on international transactions. Expected crossings provide additional information beyond the foreign-demand share.

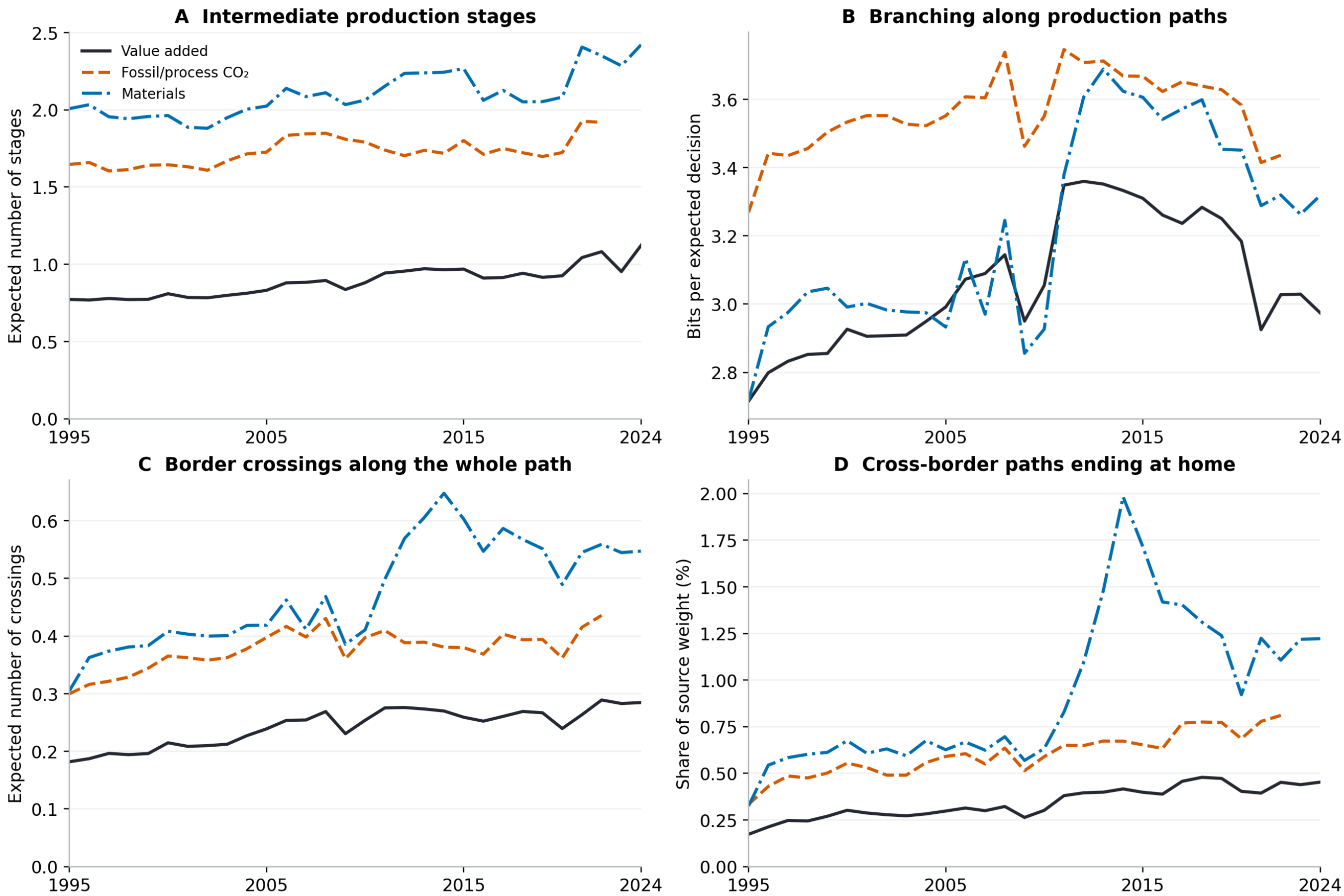

Figure 2. Distinct dimensions of production-path complexity. The same full-resolution network is weighted by positive value added, industrial carbon dioxide or used material extraction. Panels compare expected intermediate stages, branching entropy per expected decision, expected regional crossings and the probability of leaving the source region before returning to its final demand. These are positive-final-demand-closure diagnostics; historical signed-account shares are reported separately. Colours and line styles remain consistent across panels.

4.4 Source redistribution affected economic and environmental shares differently

The increase in the value-added share serving final demand abroad is almost entirely associated with changes in the destination relationships of active source accounts. For the positive-value-added weights used in the decomposition, their routing contribution is 6.51 percentage points between 1995 and 2022. The source-composition contribution is approximately 0.02 points, with a small contribution from entry and exit. Thus the redistribution of positive income weights across sources has little net endpoint effect on the global destination share, although it matters for individual countries and for intermediate dates.

Carbon shows a different weighting effect. Changes in account-level destination relationships contribute 10.64 percentage points, partly offset by a source-composition contribution of -2.90 points. Sources gaining carbon weight were, in combination, less dependent on foreign final demand than the routing contribution alone would suggest. For materials, source composition contributes little compared with the increase associated with destination relationships. Figure 3 places these results alongside value added so that the environmental decomposition is not interpreted without its economic counterpart.

The historical stage decomposition is more sensitive to the endpoint. Value-added-weighted stages increase through both 2019 and 2022, but the account-relationship contribution is much larger in the latter comparison. For carbon and materials, the routing contribution to stage length is negative through 2019 and positive through 2022. These sign changes concern different time intervals, not failed numerical reconciliation. They show why an exact decomposition still requires a defensible period and an explicit statement of what its components measure.

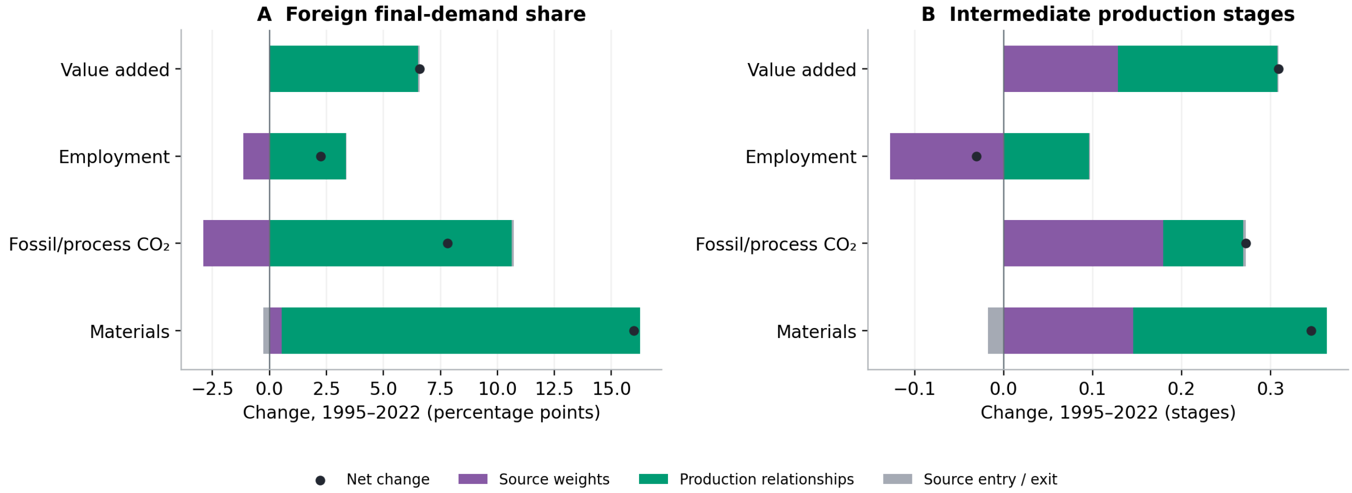

Figure 3. Decomposition of historical destination shares and intermediate-stage means, 1995–2022. Source composition is the distribution of positive value added or traceable industrial pressure across originating accounts. Routing is the account-level destination share or stage measure. Entry and exit are reported separately. Points show the net change; all components use the symmetric endpoint convention. The destination panel reports percentage points.

4.5 Country profiles reveal two main gradients

The first two principal components account for 75.5% of variation in the eight standardised country-profile features. The first component accounts for 49.7% and primarily associates greater dependence on foreign final demand with greater branching of income-weighted production paths. Carbon and material relative production intensities enter with the opposite sign. The dominant pattern therefore links international demand relationships and network branching with lower environmental source shares relative to economic value, rather than simply separating larger from smaller economies.

The second component adds a distinct gradient. Higher relative carbon and material intensities and longer income-weighted paths are associated with a lower service value-added share. This distinguishes production and resource profiles that a single integration measure would combine. The component is not a development ranking: resource endowments, domestic market size, industrial specialisation and prices can all influence a country's location. Loadings identify covariance among the selected characteristics, not the direction of influence among them.

Country trajectories illustrate why a continuous representation is useful. China and India move along the first component while remaining on a different part of the second gradient from the United States and Japan. Germany moves towards the more internationally connected part of the profile space. Norway combines substantial international connections with a resource-related position on the second component. Figure 4 shows selected trajectories and all feature loadings; the full country-year scores permit comparisons without selecting cases solely for their visual separation.

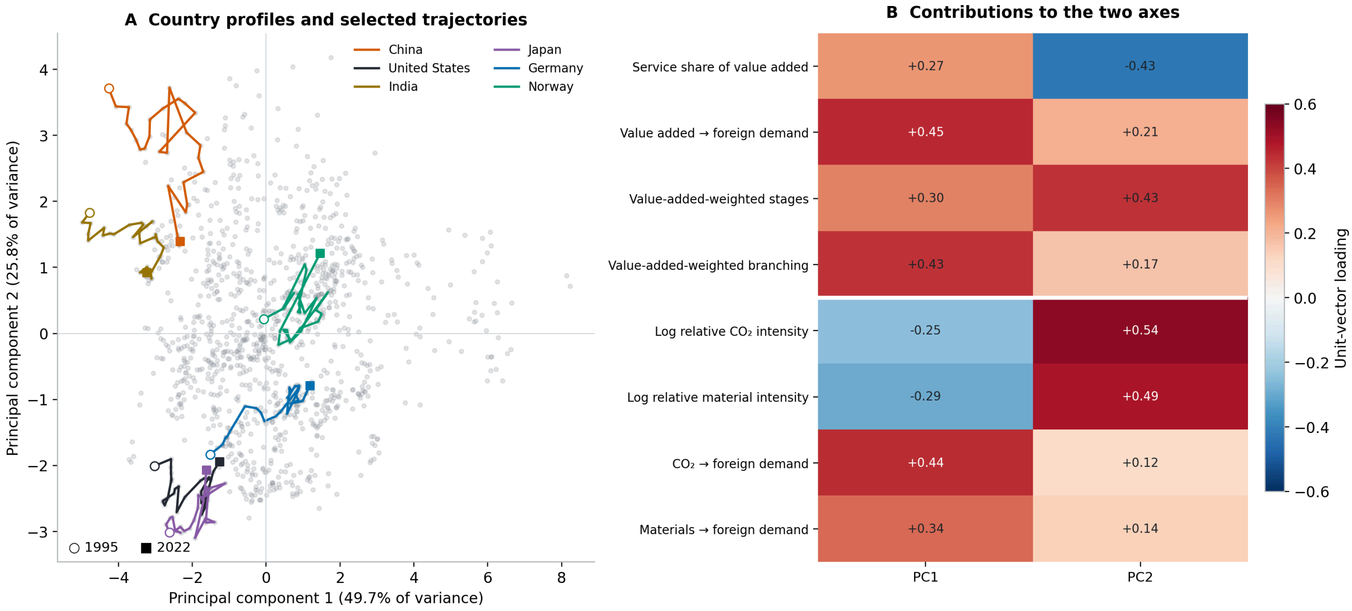

Figure 4. Principal component analysis of 1,232 country-year profiles. A: selected country trajectories from 1995 to 2022 within the full cloud of observations. B: loadings of the four economic and four environmental features on the first two components. PCA uses all eight standardised features; component signs are fixed only for reproducible display. The country trajectories are descriptive and do not imply causal paths.

The best admissible clustering solution contains two groups, but its silhouette score of 0.269 indicates appreciable overlap. Their centroids mainly distinguish lower and higher dependence on foreign final demand. The higher-dependence group also has more income-weighted branching, longer income-weighted paths, a larger service share and lower relative carbon and material production intensities. Because the groups compress continuous gradients, these properties should be read as average profiles rather than rules applying to every member.

The number of countries assigned to the higher-dependence group rises from nine in 1995 to thirty in 2022. This gives a compact description of a widespread movement in the selected features, but a change of label does not identify an abrupt economic transition. Some countries cross a fitted boundary as their profile changes gradually. Others retain their label despite substantial movement within the same broad group. The PCA trajectories and assignment margins are therefore retained alongside the clustering rather than replaced by a categorical map.

4.6 Stability favours gradients over rigid country types

The cluster partition is reasonably stable to the omitted-period and scaling checks, with adjusted Rand indices between approximately 0.89 and 0.95. Stability is weaker when countries are resampled: the median index is 0.78 and its tenth percentile is 0.54. The differences show that the typology depends more on which country histories are represented than on the tested time blocks. These are reasons to retain the grouping as an exploratory summary and avoid treating it as an externally validated classification.

The first two components explain only 43.9% of variation when PCA is fitted to annual within-country changes. The compact level representation therefore does not extend unchanged to annual movements. The observed level share of 75.5% also exceeds the 69.9% upper reference quantile from independently circular-shifted feature histories, suggesting that contemporaneous alignment contributes information beyond preserved marginal histories. Because this reference retains country means and alters temporal boundaries, it supports no formal causal or independence claim.

The accounting and path calculations pass their numerical checks. The maximum path row-probability discrepancy is approximately 1.30×10^{-14} and the maximum reward-equation residual 6.64×10^{-14}. Negative value added accounts for at most 0.108% of absolute value added, and positive demand raises total required output by at most 0.718% relative to the original accounts. These checks establish that the probability construction is numerically coherent and that its global closure adjustment is small. They do not eliminate source-data uncertainty or ensure that every country-level difference is equally small.

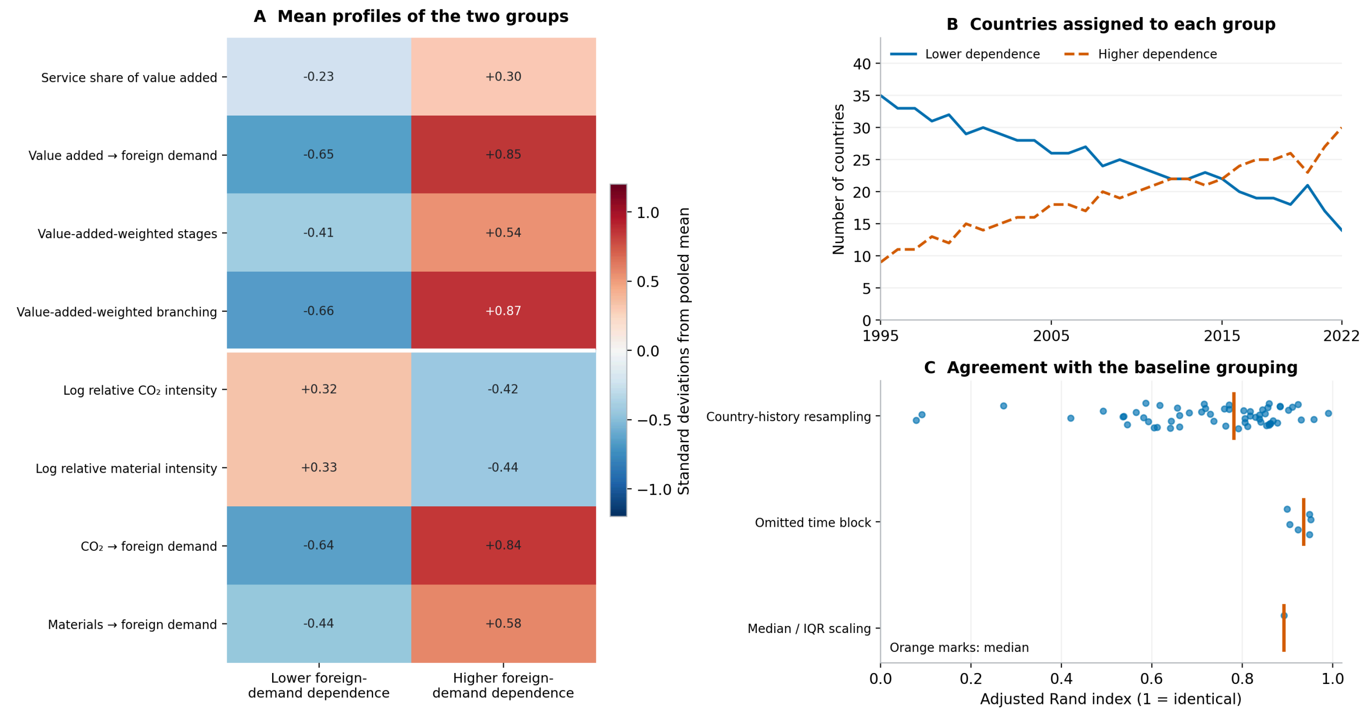

Figure 5. Exploratory cluster profiles and stability. Standardised centroids show the two fitted profiles across the same four economic and four environmental features. Membership counts describe the selected named-country sample. Stability statistics compare refitted partitions with the baseline and are not confidence intervals for country types. The modest silhouette score and dependence on the country sample favour interpretation as overlapping profiles.

5. Discussion

5.1 Economic transformation and environmental dependence

The economic results qualify two familiar descriptions of global structural change. First, a decline in manufacturing's global value-added share does not imply an equally large expansion of services. The relatively stable broad service share conceals changes within services and in extraction, utilities, construction and other activities. Second, the redistribution of economic value towards emerging economies does not guarantee persistent geographical diversification. The effective number of value-added source regions increased and then declined. Describing these movements separately gives a more precise account of the economy than a single transition from industry to services or from national to global production.

The production-network results add a dimension that sector shares cannot supply. A larger fraction of value added was generated for final demand outside its source region, and income-weighted paths became longer. These changes mean that the economic link between income generation and final expenditure involved more intermediate relationships and more geographically separated demand. They do not establish that domestic production became less valuable or that longer chains improved productivity. They describe how the recorded income was connected to purchases, which is relevant to understanding economic interdependence even without a causal productivity estimate.

Environmental pressure changed along a different set of weights. Carbon and material sources became more concentrated while value-added industry diversity increased modestly. Their paths remained further upstream than the average income-generating activity, and larger shares served foreign final demand. This is not a contradiction between the economic and environmental accounts. Both apply the same production relationships to source quantities distributed differently across industries and countries. The comparison identifies the importance of retaining both quantities rather than assuming that a monetary description adequately represents environmental dependence.

The near-flat land inventory reinforces the need to distinguish environmental dimensions. An area measure can remain stable while material extraction and energy use rise, because changes in use intensity and resource throughput need not expand the recorded area. Equally, a larger service share can coexist with increased resource extraction because services require infrastructure, equipment and energy. These observations do not establish the environmental merit of any individual sector. They show why a structural account should pair economic composition with physical inventories and the production relationships connecting them.

5.2 What complexity adds to the explanation

The path analysis changes the interpretation of an apparently simple trend. Material production length was nearly unchanged between 1995 and 2019, yet branching per expected decision increased. The system could offer more alternative production paths without adding many stages to the average path. After 2019, the mean length increased while branching per decision fell. Calling both periods an increase in complexity would erase their different mechanisms. Describing length, dispersion and branching separately preserves information that a weighted composite index would suppress.

The same distinction applies to international production. A final-demand share records the relationship between a source and its ultimate purchaser's region. An expected crossing count records the intermediate international relationships along the path. Both are economically meaningful, but they answer different questions. The increase in repeated crossings indicates more international transitions per path that crosses at least once. The additional domestic-return measure identifies dependence on foreign production relationships that remains outside a simple foreign-final-demand share. Its small size in the present results should not be inflated into a dominant explanation, but neither should it be assumed to be zero.

These measures also identify limits of the common resilience interpretation of network statistics. Greater branching could represent a diversified set of relationships, but the IO table does not show whether one supplier can replace another during disruption. Several branches may depend on the same technology, resource, infrastructure or financial condition. A long path may have redundant capacity, whereas a short path may contain a unique critical input. The present statistics describe the arrangement of recorded relationships. Assessing vulnerability requires additional assumptions about capacities, substitutability, inventories and the dynamics of adjustment. Path entropy therefore requires a narrow interpretation.

5.3 What the exploratory profiles establish

The PCA results indicate that economic composition, foreign final-demand dependence and environmental production intensities form related but non-identical gradients across countries. The first component associates greater foreign-demand dependence and income-weighted branching with lower environmental source shares relative to value added. The second separates more resource- and production-intensive profiles from more service-oriented profiles. Their coexistence helps explain why two countries with similar international integration can occupy different economic and environmental positions.

The association between integration and lower relative production intensity must not be interpreted as evidence that integration reduces global pressure. A country can have a low domestic environmental source share partly because it imports products whose upstream pressure occurs elsewhere. Its current-price value-added share also depends on the prices and income generated by its activities. The environmental features in the PCA describe production sources, while the destination shares describe the final demand connected to those sources. They do not constitute a complete consumption-footprint comparison across countries. The global inventory results independently show that pressure expanded during the period.

The cluster results provide a convenient summary of this feature space, but the moderate silhouette and country-resampling sensitivity argue against a rigid typology. Two groups capture a broad difference in foreign-demand dependence; they do not establish two natural kinds of economy. Continuous scores, original features and assignment margins are more informative for countries near the boundary. The movement from nine to thirty countries in the higher-dependence group describes how the sample occupies a fixed fitted partition. It is not a count of countries crossing an externally defined development threshold.

The annual-change comparison further limits interpretation. Most variation in the level features can be represented by two components, but annual movements are less compressible. Some prominent associations therefore reflect persistent country differences and gradual change rather than a common mechanism operating each year. This is precisely why the analysis includes both level profiles and within-country changes. A visually compact PCA plot should not be allowed to imply that structural evolution follows a single trajectory or that a country's future path can be predicted from its position.

5.4 Implications for empirical and policy research

The most direct implication is methodological: economic income generation and environmental pressure should be followed through the same detailed production network, with their weighting and destination conventions stated. A production-based environmental inventory, an income share and a final-demand footprint are not rival estimates of the same quantity. They locate different parts of the relationship between activity and demand. Their joint interpretation can identify questions that remain hidden when research starts from only territorial emissions or only monetary sector shares.

For policy analysis, the results identify where additional specification is needed. A high share of income or pressure serving foreign final demand raises questions about the coordination of producers, purchasers and public authorities across regions. Repeated border crossings raise questions about intermediates and returning production, not only final imported goods. Concentrated environmental sources identify activities where direct process changes might matter substantially. None of these findings supplies the effect of a policy without specifying the production service to be maintained, the available alternatives and the responses of demand and supply.

Investment illustrates this boundary. In the underlying demand accounts, the carbon share attributed from production sources to gross fixed capital formation increased, whereas investment's material share changed less. That observation directs attention to buildings, machinery and infrastructure alongside current consumption. It does not determine whether the associated assets increase or reduce later operating pressure. A credible investment scenario would need stock turnover, service lives, operating technologies and the production requirements of replacement assets. Treating capital formation as final demand is appropriate for the annual accounting comparison but incomplete for evaluating a transition over time.

Likewise, the difference between income- and employment-weighted production paths suggests questions about the distribution of economic gains and adjustment burdens. It cannot identify workers at risk. Occupational skills, wages, household dependence and local labour markets are absent from the global people counts. A policy that changes a carbon-intensive source can affect economic activity elsewhere through its customers and suppliers. Linking the descriptive network to feasible technology and labour-market adjustments is therefore a separate modelling task, requiring its own assumptions and evidence.

5.5 Limits and further tests

This study uses one estimated MRIO system. Harmonised tables contain statistical information, interpolation, balancing and modelling, so every annual difference should not be read as a directly observed change in firm behaviour. The sharp late movements in some path measures require particular care. The pre-pandemic comparison and annual series reveal their importance; they do not explain the underlying database changes. The separate carbon inventory supports the broad growth direction and approximate magnitude, but it cannot establish the correctness of detailed income or environmental routes.

Monetary interpretation remains another limit. Current-price value added is appropriate for describing the distribution of recorded income, but it cannot identify real growth or technical efficiency. Consistent account-level revaluation leaves the conditional production-path probabilities unchanged, yet economic source weights and relative environmental intensities can still change. More detailed price and product information would be needed to separate these effects. The study therefore does not relabel a declining pressure-per-euro ratio as decarbonisation of a physical production process.

The positive-demand closure is small in aggregate but remains a model choice. It supplies nonnegative probabilities while preserving the original input coefficients, and its differences from historical signed measures are retained in the outputs. Small worldwide adjustments can conceal larger local effects. For this reason, country-profile interpretation should be checked against the original feature values and annual audit records when a particular country is the focus. The global checks establish coherence; they do not excuse ignoring the boundary of an individual result.

Future replication should test the principal economic, environmental and path findings in another MRIO with harmonised concepts and a clearly stated period. Additional environmental characterisation should precede claims about water stress or biodiversity damage. Country-profile extensions should compare theoretically motivated feature sets and external economic evidence before naming development regimes. These tests would address remaining empirical questions; adding more complex machine-learning algorithms alone would not resolve them.

6. Conclusion

Global structural change involved a redistribution of economic value, a decline in manufacturing's share, greater dependence of income generation on foreign final demand, and changes in production paths. The broad service share increased only modestly, while economic geographical diversity rose and then receded. Carbon and material inventories expanded and became more concentrated at their sources. Economic diversification therefore did not imply declining environmental dependence.

The added path measures distinguish changes that average production length alone misses. Longer paths, more branching and repeated regional crossings followed different histories. Exploratory machine learning summarises related country characteristics into interpretable gradients, while stability checks limit the case for fixed country types. Together, the findings support a structural account that treats economic value and environmental pressure with equal attention and separates measured relationships from claims about feasible alternatives.

References

Andrew, R. M., & Peters, G. P. (2025). GCP fossil CO2 emissions dataset, 2025v15 [Data set]. Zenodo. Source

Antràs, P., Chor, D., Fally, T., & Hillberry, R. (2012). Measuring the upstreamness of production and trade flows. American Economic Review, 102(3), 412–416. Source

Barney, J. B. (2018). Editor's comments: Positioning a theory paper for publication. Academy of Management Review, 43(3), 345–348. Source

Crameri, F., Shephard, G. E., & Heron, P. J. (2020). The misuse of colour in science communication. Nature Communications, 11, 5444. Source

Dietzenbacher, E., & Romero, I. (2007). Production chains in an interregional framework: Identification by means of average propagation lengths. International Regional Science Review, 30(4), 362–383. Source

Dietzenbacher, E., van Burken, B., & Kondo, Y. (2019). Hypothetical extractions from a global perspective. Economic Systems Research, 31(4), 505–519. Source

EXIOBASE consortium. (2026). EXIOBASE 3, release 3.10.2 [Data set, release metadata and documentation]. Zenodo. Source

Ekroot, L., & Cover, T. M. (1993). The entropy of Markov trajectories. IEEE Transactions on Information Theory, 39(4), 1418–1421. Source

Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: a review and recent developments. Philosophical Transactions of the Royal Society A, 374(2065), 20150202. Source

Kostoska, O., Stojkoski, V., & Kocarev, L. (2020). On the structure of the world economy: An absorbing Markov chain approach. Entropy, 22(4), 482. Source

OECD. (2026). Inter-country input-output tables: 2025 edition, revised January 2026 [Database documentation]. Source

Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830. Source

Peters, G. P., Minx, J. C., Weber, C. L., & Edenhofer, O. (2011). Growth in emission transfers via international trade from 1990 to 2008. Proceedings of the National Academy of Sciences, 108(21), 8903–8908. Source

Resource Use. (2026). Data and models: GLORIA [Provider documentation, accessed 6 September 2026]. Source

Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics, 20, 53–65. Source

Stadler, K., Wood, R., Bulavskaya, T., Södersten, C.-J., Simas, M., Schmidt, S., Usubiaga, A., Acosta-Fernández, J., Kuenen, J., Bruckner, M., Giljum, S., Lutter, S., Merciai, S., Schmidt, J. H., Theurl, M. C., Plutzar, C., Kastner, T., Eisenmenger, N., Erb, K.-H., de Koning, A., & Tukker, A. (2018). EXIOBASE 3: Developing a time series of detailed environmentally extended multi-regional input-output tables. Journal of Industrial Ecology, 22(3), 502–515. Source

Tufte, E. R. (2001). The visual display of quantitative information (2nd ed.). Graphics Press. Source

UNEP International Resource Panel. (2024). Global resources outlook 2024: Bend the trend—Pathways to a liveable planet as resource use spikes. Source

University of Groningen. (2016). WIOD 2016 release [Database documentation]. Source

Wiedmann, T. O., Schandl, H., Lenzen, M., Moran, D., Suh, S., West, J., & Kanemoto, K. (2015). The material footprint of nations. Proceedings of the National Academy of Sciences, 112(20), 6271–6276. Source

Wood, R., & Lenzen, M. (2009). Aggregate measures of complex economic structure and evolution: A review and case study. Journal of Industrial Ecology, 13(2), 264–283. Source

Wood, R., Moran, D. D., Rodrigues, J. F. D., & Stadler, K. (2019). Variation in trends of consumption based carbon accounts. Scientific Data, 6, 99. Source

World MRIO. (2026). Eora26 [Provider documentation, accessed 6 September 2026]. Source

Xu, Y., & Dietzenbacher, E. (2014). A structural decomposition analysis of the emissions embodied in trade. Ecological Economics, 101, 10–20. Source